

NJORD - Nav2 & GPS Report

Summary: Concise summary of work on Nav2 and GPS integration for reliable, collision-free navigation

Key Objectives:

- Robust Nav2 planning and safe local control
- GPS integration for global waypoint following and coarse pose updates

Work Completed:

- Tuned global/local planners, costmaps, and recovery behaviors
- Implemented GPS fusion with odometry (EKF) for improved global localization
- Validated in sim and on hardware; created reproducible launch files

Results: >95% collision-free runs in test course; primary failures due to GPS outages

Next Steps: Improve GPS robustness (multi-constellation/RTK), final Nav2 tuning and deployment

Contact: NODE Engineering Club - Nav2 Team